

DA3408 AI Operations Assignment 1

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1.1

(a) **Entanglement (CACE):** All the features feed the same model, so changing the delivery time rounding shifted the whole model and made the unrelated favorite-restaurants feature worse.

(b) **Undeclared Consumers:** The marketing team is silently reading the model's output table and the ML team does not know, so any change can break their dashboard.

(c) **Configuration and Glue-Code Debt:** The pipeline is 14 undocumented shell scripts with no orchestration and the settings are scattered with no single source of truth.

1.2

For (c), I would add **MLflow tracking** to the pipeline, which is the direct fix for configuration debt. Every run logs its parameters and metrics in one place with `mlflow.log_param` and `mlflow.log_metric` instead of the settings staying hidden inside 14 scripts. With an `environment.yml` to pin the versions, there is one source of truth and it is clear which settings gave which result.

2

I ran six MLP experiments on MNIST by changing the learning rate and the batch size. The best run was `mlp-lr-0.001-batch-32` with validation accuracy **0.9775**. I chose it because validation accuracy shows how the model does on unseen data, not training loss.

There is some overfitting. The run `mlp-lr-0.001-batch-128` had the lowest training loss (**0.0081**) but its validation accuracy was only **0.9733**. For learning rate 0.001, training loss went down as batch size went up, but validation accuracy went down at the same time.

The **learning rate** has the larger effect. Runs with learning rate 0.001 gave accuracy between 0.9733 and 0.9775, and runs with 0.01 gave between 0.9594 and 0.9689, so the two groups do not overlap. Batch size changed accuracy by only about 0.004 inside a group. The logging code is in `Question 2/Assignment1_Q2.ipynb`.

3

I made a CSV of the file names with 1800 rows and a header, ran `dvc init` and `dvc add`, pushed to an SSH remote and committed it as **v1**. I then added the rows from `new-labels.zip` to get 2801 lines and committed that as **v2**. For the rollback I ran `git checkout v1` and `dvc checkout`, after which `wc -l` gave **1801 lines**, matching v1 exactly. The output is in `Question 3/q3_3.png`.

4

As Partner B I cloned Partner A's repository, checked out commit `ff5e88c`, built the environment from `environment.yml`, ran `dvc checkout` and reran the training script unchanged. I got `val_accuracy = 0.9639` and `val_f1_macro = 0.9635`, the same as Partner A, so it matched within the tolerance of **0.001**.